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CSE 1205 — Mid Term Suggestion Notes

CSE 1205 | Introduction to Computer Science — Mid Term Exam Preparation

A comprehensive study guide organized by question number.

Question 1: Memory Systems

1.1 Primary Memory (Main Memory)

💡 Primary memory is the computer's main internal memory that the CPU accesses directly. It is **volatile** (loses data when power is off) or **non-volatile**, and is faster but smaller than secondary memory.

Key Characteristics:

- Directly accessible by the CPU
- Faster access speed compared to secondary memory
- Limited in capacity
- More expensive per byte
- Includes **RAM** and **ROM**

1.2 Secondary Memory (Auxiliary Memory)

💾 Secondary memory is **non-volatile** storage used for permanent data retention. The CPU cannot access it directly — data must first be loaded into primary memory.

Key Characteristics:

- Non-volatile (data persists without power)
- Larger storage capacity
- Slower access speed
- Cheaper per byte
- Examples: Hard Disk Drive (HDD), Solid State Drive (SSD), CD/DVD, USB Flash Drive, Magnetic Tape

Comparison: Primary vs Secondary Memory

Feature	Primary Memory	Secondary Memory
Volatility	Volatile (RAM) / Non-volatile (ROM)	Non-volatile
Speed	Very fast	Slower
Capacity	Small (GBs)	Large (TBs)
Cost	Expensive per byte	Cheap per byte
CPU Access	Direct	Indirect (via primary memory)
Examples	RAM, ROM, Cache	HDD, SSD, USB, DVD

1.3 RAM (Random Access Memory)

⚡ RAM is **volatile** primary memory used to store data and instructions currently being processed by the CPU. It allows **read and write** operations.

Types of RAM:

SRAM (Static RAM)

- Uses **flip-flop circuits** to store data
- **Faster** and does **not need refreshing**
- More **expensive**, lower density
- Used in **cache memory**

DRAM (Dynamic RAM)

- Uses **capacitors** to store data
- **Slower** and **needs periodic refreshing**
- **Cheaper**, higher density
- Used as **main memory** in computers

Feature	SRAM	DRAM
Storage Element	Flip-flops	Capacitors
Speed	Faster	Slower
Refresh Needed	No	Yes
Cost	Expensive	Cheaper
Density	Low	High
Use	Cache memory	Main memory

1.4 ROM (Read-Only Memory)

🔒 ROM is **non-volatile** primary memory that stores permanent instructions (e.g., BIOS/firmware). Data is written during manufacturing and **cannot be easily modified**.

Types of ROM:

1. **PROM (Programmable ROM)** — Can be programmed **once** after manufacturing using a special device.
2. **EPROM (Erasable Programmable ROM)** — Can be erased using **ultraviolet (UV) light** and reprogrammed.
3. **EEPROM (Electrically Erasable Programmable ROM)** — Can be erased and reprogrammed **electrically**, byte by byte.
4. **Flash Memory** — A type of EEPROM that can be erased and written in **blocks**; used in USB drives, SSDs.

1.5 Cache Memory

🚀 Cache is a **very fast, small** memory located between the CPU and main memory (RAM). It stores **frequently accessed data and instructions** to speed up processing.

How Cache Works:

- When the CPU needs data, it first checks the cache (**cache hit**).
- If not found (**cache miss**), it fetches from RAM and copies to cache.
- Principle: **Locality of Reference** — programs tend to access the same data/instructions repeatedly.

Levels of Cache:

Level	Location	Size	Speed
L1 Cache	Inside CPU core	Smallest (32–128 KB)	Fastest
L2 Cache	Inside or near CPU	Medium (256 KB – 1 MB)	Fast
L3 Cache	Shared among cores	Largest (2–64 MB)	Slower than L1/L2

1.6 Memory Hierarchy

📊 The memory hierarchy organizes storage by **speed, cost, and capacity**. Faster memory is smaller and more expensive; slower memory is larger and cheaper.

Memory Hierarchy (Top = Fastest, Bottom = Slowest):

1. **CPU Registers** — Fastest, smallest, inside CPU
2. **L1 Cache** — Very fast, very small
3. **L2 Cache** — Fast, small
4. **L3 Cache** — Moderate speed, moderate size
5. **RAM (Main Memory)** — Moderate speed, moderate capacity
6. **SSD / HDD (Secondary Storage)** — Slow, large capacity
7. **Magnetic Tape / Optical Discs (Tertiary Storage)** — Slowest, largest, cheapest

Key Principle:

> As we move **down** the hierarchy:
> → **Speed decreases**
> → **Capacity increases**
> → **Cost per byte decreases**
> → **Access time increases**

Question 2(a) + 2(b) + 4: Number Systems & Computer Types

2.1 Types of Computers

🖥️ Computers are classified based on **size, purpose, and data handling**.

A. Based on Size

1. **Supercomputer**
   - Most powerful; used for weather forecasting, scientific simulations, nuclear research
   - Examples: IBM Summit, Fugaku
2. **Mainframe Computer**
   - Large, supports thousands of users simultaneously
   - Used in banking, airlines, government
   - Examples: IBM z15
3. **Minicomputer (Mid-range)**
   - Smaller than mainframes; multi-user
   - Used in small businesses and departments
   - Example: PDP-11
4. **Microcomputer (Personal Computer)**
   - Single-user; most common

- Includes desktops, laptops, tablets, smartphones

5. **Workstation**
   - High-performance microcomputer for engineering, design, scientific work

B. Based on Purpose

1. **General-Purpose Computer** — Can perform a wide variety of tasks (e.g., desktops)
2. **Special-Purpose Computer** — Designed for a specific task (e.g., traffic signal controller, ATM)

C. Based on Data Handling

1. **Analog Computer** — Processes continuous data (e.g., thermometer, speedometer)
2. **Digital Computer** — Processes discrete/binary data (e.g., PCs, laptops)
3. **Hybrid Computer** — Combines analog and digital (e.g., hospital ICU monitors)

2.2 Social Impact of Computers

🌍 Computers have transformed virtually every aspect of modern society — both positively and negatively.

Positive Impacts

- **Education** — E-learning, online courses, digital libraries, virtual classrooms
- **Healthcare** — Computerized diagnostics, telemedicine, electronic health records, robotic surgery
- **Communication** — Email, social media, video conferencing, instant messaging
- **Business & Economy** — E-commerce, online banking, automated accounting, digital marketing
- **Entertainment** — Video games, streaming, digital art, virtual reality

- **Science & Research** — Simulations, data analysis, genome mapping
- **Government** — E-governance, digital ID systems, online tax filing

Negative Impacts

- **Unemployment** — Automation replacing human jobs
- **Privacy Concerns** — Data breaches, surveillance, identity theft
- **Cybercrime** — Hacking, phishing, ransomware
- **Health Issues** — Eye strain, poor posture, screen addiction
- **Digital Divide** — Unequal access to technology across socioeconomic groups
- **Social Isolation** — Reduced face-to-face interaction
- **Misinformation** — Spread of fake news and propaganda

2.3 Number System Conversions

> 🔢 Computers use **binary (base-2)**. Understanding conversions between **Decimal (base-10), Binary (base-2), Octal (base-8), and Hexadecimal (base-16)** is essential.

Decimal to Binary (Integer)

Method: Repeated division by 2, record remainders bottom-to-top.

Example: Convert **$(25)_{10}$** to Binary

Division	Quotient	Remainder
25 ÷ 2	12	1
12 ÷ 2	6	0
6 ÷ 2	3	0
3 ÷ 2	1	1
1 ÷ 2	0	1

Result: $(25)_{10} = (11001)_2$ ← Read remainders bottom to top

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Octal to Hexadecimal

Method: Octal → Binary → Hexadecimal

Step 1: Convert each octal digit to **3-bit binary**

Step 2: Group binary digits into sets of **4 from right**

Step 3: Convert each group to hexadecimal

Example: Convert **$(725)_8$** to Hexadecimal

- 7 = 111, 2 = 010, 5 = 101
- Binary: **111 010 101**
- Regroup into 4 bits: **0001 1101 0101**
- 0001 = 1, 1101 = D, 0101 = 5
- **Result: $(725)_8 = (1D5)_{16}$**

Binary to Octal

Method: Group binary digits into sets of **3 from right**, convert each group.

Example: Convert **$(110101011)_2$** to Octal

- Group: **110 101 011**
- 110 = 6, 101 = 5, 011 = 3
- **Result: $(110101011)_2 = (653)_8$**

Hexadecimal to Decimal

Method: Multiply each digit by its positional power of 16, then sum.

Example: Convert **$(2A3)_{16}$** to Decimal

- $2 \times 16^2 = 2 \times 256 = 512$
- A (10) $\times 16^1 = 10 \times 16 = 160$
- $3 \times 16^0 = 3 \times 1 = 3$
- **Result: $(2A3)_{16} = 512 + 160 + 3 = (675)_{10}$**

2.4 Octal ↔ Hexadecimal Using BCD (Binary Coded Representation)

📐 **BCD-style approach:** Each octal digit maps to **3 binary bits**, each hex digit maps to **4 binary bits**. Convert by going through binary as an intermediate.

Octal to Hex (via Binary)

Example: $(374)_8$ → Hex

1. Octal to 3-bit binary: 3 = 011, 7 = 111, 4 = 100
2. Combine: **011 111 100**
3. Regroup into 4: **0 0111 1110 0** → pad: **0000 1111 1100**
4. 0000 = 0, 1111 = F, 1100 = C
5. **Result: $(374)_8 = (FC)_{16}$**

Hex to Octal (via Binary)

Example: $(1F4)_{16}$ → Octal

1. Hex to 4-bit binary: 1 = 0001, F = 1111, 4 = 0100
2. Combine: **0001 1111 0100**
3. Regroup into 3: **000 111 110 100**
4. 000 = 0, 111 = 7, 110 = 6, 100 = 4
5. **Result: $(1F4)_{16} = (764)_8$**

2.5 Binary Multiplication

✖️ Binary multiplication follows the same principle as decimal multiplication but with only 0s and 1s.

Rules:

- 0 × 0 = 0
- 0 × 1 = 0
- 1 × 0 = 0

- 1 × 1 = 1

Example: Multiply **$(1101)_2 \times (1011)_2$**

```
      1 1 0 1   (Multiplicand = 13)
    × 1 0 1 1   (Multiplier = 11)
    ---------
      1 1 0 1   (1101 × 1)
    1 1 0 1 0   (1101 × 1, shifted left 1)
  0 0 0 0 0 0   (1101 × 0, shifted left 2)
1 1 0 1 0 0 0   (1101 × 1, shifted left 3)
---------------
1 0 0 0 1 1 1 1  = (143)10 ✓ (13 × 11 = 143)
```

2.6 Binary Division

➗ Binary division works like long division in decimal, using **repeated subtraction and shifting**.

Example: Divide **$(10110)_2 \div (110)_2$** → i.e., 22 ÷ 6

```
        0 0 1 1  ← Quotient (= 3)
      --------
110 ) 1 0 1 1 0
      - 0 0 0     (110 doesn't fit into 1)
      -------
        1 0 1     (110 doesn't fit into 10, fits into 101?
No.)
        Bring down: 1011
      - 1 1 0
      -------
        1 0 1 0
      - 1 1 0
      -------
        1 0 0  ← Remainder (= 4)
```

Result: $(10110)_2 \div (110)_2$ = Quotient $(11)_2$, Remainder $(100)_2$

Verification: 22 ÷ 6 = 3 remainder 4 ✓

2.7 Binary Subtraction with 2's Complement

➖ **2's Complement** is the most common method for representing negative numbers and performing subtraction in computers.

Formula: 2's Complement = 1's Complement + 1
1's Complement = Flip all bits (0 → 1, 1 → 0)

Steps for A – B using 2's Complement:

1. Find the **1's complement** of B (flip all bits)
2. Add **1** to get the **2's complement** of B
3. Add A + (2's complement of B)
4. If there is a **carry out** → result is **positive** (discard carry)
5. If there is **no carry** → result is **negative** (take 2's complement of result)

Example 1: $(1010)_2 - (0111)_2$ → i.e., 10 – 7

1. 1's complement of 0111 = **1000**
2. 2's complement = 1000 + 1 = **1001**
3. Add: 1010 + 1001 = **1 0011**
4. Carry = 1 → Result is **positive**, discard carry
5. **Result: $(0011)_2 = (3)_{10}$** ✓

Example 2: $(0100)_2 - (0111)_2$ → i.e., 4 – 7

1. 1's complement of 0111 = **1000**
2. 2's complement = 1000 + 1 = **1001**
3. Add: 0100 + 1001 = **1101** (no carry beyond 4 bits)
4. No carry → Result is **negative**
5. Take 2's complement of 1101: 0010 + 1 = **0011**
6. **Result: $-(0011)_2 = -(3)_{10}$** ✓

Question 3: Input/Output Devices

3.1 Keyboard Layout

⌨️ A keyboard is the most common **input device** for entering text, numbers, and commands.

Major Keyboard Layouts

1. QWERTY Layout

- Most widely used worldwide
- Named after the first 6 letters on the top letter row: Q-W-E-R-T-Y
- Designed in the 1870s by **Christopher Latham Sholes** for typewriters
- Designed to **reduce jamming** in mechanical typewriters by separating commonly used letter pairs

2. DVORAK Layout

- Designed by **Dr. August Dvorak** in 1936
- Optimized for **speed and efficiency** — most common letters are on the home row
- Less popular due to QWERTY dominance

3. AZERTY Layout

- Used primarily in **France and Belgium**
- Variation of QWERTY adapted for French language characters

Sections of a Standard Keyboard

Section	Keys	Purpose
Alphanumeric Keys	A-Z, 0-9, symbols	Typing text and numbers
Function Keys	F1 – F12	Shortcuts for specific operations

Section	Keys	Purpose
Navigation Keys	Arrow keys, Home, End, PgUp, PgDn	Moving cursor and scrolling
Numeric Keypad	0-9, +, −, *, /, Enter	Quick numeric data entry
Modifier Keys	Shift, Ctrl, Alt, Caps Lock	Modify behavior of other keys
Special Keys	Enter, Backspace, Tab, Esc, Space	Control and editing functions

3.2 Mouse Types

🖱️ A mouse is a **pointing input device** that detects two-dimensional motion relative to a surface and translates it into cursor movement on screen.

Types of Mouse

1. Mechanical Mouse (Ball Mouse)

- Contains a **rubber ball** on the bottom
- Ball rolls when moved, and internal **rollers** detect direction
- Requires a **mouse pad** for proper functioning
- **Disadvantage:** Collects dust, requires regular cleaning, less precise

2. Optical Mouse

- Uses an **LED light** and **optical sensor (photodiode)** to detect movement
- No moving parts → more reliable
- Works on most surfaces (not glass)
- Most **commonly used** today

3. Laser Mouse

- Uses a **laser beam** instead of LED for tracking
- **Higher precision** (higher DPI) than optical mouse
- Works on more surface types including glass
- Preferred by **gamers and designers**

4. Wireless Mouse

- Connects via **Bluetooth** or **RF (Radio Frequency) receiver (USB dongle)**
- No cable → more freedom of movement
- Requires **batteries or charging**

5. Trackball Mouse

- Has a **stationary ball on top** that the user rotates with thumb/fingers
- The body doesn't move → requires **less desk space**
- Used in specialized workstations

6. Gaming Mouse

- High **DPI (dots per inch)** for precision
- **Programmable buttons**, adjustable weights
- Ergonomic design for extended use

3.3 Monitor Types and Comparison

🖥️ A monitor (display screen) is the primary **output device** for visual information.

Types of Monitors

1. CRT (Cathode Ray Tube)

- Oldest technology
- Uses **electron beams** to illuminate phosphor dots on screen
- **Bulky and heavy**, high power consumption
- Good color reproduction for its time
- Now largely **obsolete**

2. LCD (Liquid Crystal Display)

- Uses **liquid crystals** between two layers of polarized glass
- Requires a **backlight** (usually LED or CCFL)

- **Thin, lightweight, energy-efficient**
- Most common in laptops and budget monitors

3. LED (Light-Emitting Diode)

- A type of LCD that uses **LED backlighting** instead of CCFL
- **Better contrast, brightness, and energy efficiency** than standard LCD
- Thinner and lighter
- Most common monitor type today

4. OLED (Organic LED)

- Each pixel produces its **own light** — no backlight needed
- **Perfect blacks** (pixels can turn completely off)
- **Excellent contrast ratio and color accuracy**
- Thinner and flexible
- **Expensive**, risk of **burn-in**

5. AMOLED (Active Matrix OLED)

- Advanced OLED with **active matrix** for faster pixel response
- Used in **smartphones and premium displays**

Comparison Table:

Feature	CRT	LCD	LED	OLED
Thickness	Very bulky	Thin	Thinner	Thinnest
Weight	Heavy	Light	Lighter	Lightest
Power Use	High	Moderate	Low	Very low
Contrast	Moderate	Moderate	Good	Excellent
Black Level	Good	Poor (backlight bleed)	Better	Perfect (true black)
Cost	Obsolete	Affordable	Moderate	Expensive
Lifespan	Long	Long	Long	Shorter (burn-in risk)

3.4 Need for Modem and Why It Is a Peripheral

📡 A **modem (Modulator-Demodulator)** is a device that converts digital signals from a computer into analog signals for transmission over telephone/cable lines, and vice versa.

What Does a Modem Do?

- **Modulation:** Converts **digital data** → **analog signal** (for sending)
- **Demodulation:** Converts **analog signal** → **digital data** (for receiving)
- Enables computers to communicate over networks that were originally designed for analog signals (telephone lines)

Why Do We Need a Modem?

1. **Internet Access** — Connects your computer to your Internet Service Provider (ISP)
2. **Data Transmission** — Enables sending/receiving data over telephone or cable lines
3. **Signal Conversion** — Bridges the gap between digital computers and analog communication infrastructure
4. **Remote Communication** — Enables email, web browsing, video calls, file transfer

Types of Modems

- **Dial-up Modem** — Uses telephone line, very slow (up to 56 Kbps)
- **DSL Modem** — Uses telephone line with higher speed (up to 100+ Mbps)
- **Cable Modem** — Uses cable TV line, high speed
- **Fiber Modem (ONT)** — Converts fiber optic signals, fastest
- **Wireless/Cellular Modem** — Uses mobile networks (3G/4G/5G)

Why Is a Modem a Peripheral Device?

✅ A **peripheral device** is any external or auxiliary device that connects to and works with a computer but is **not part of the core computer system** (CPU, motherboard, primary memory).

A modem qualifies as a peripheral because:

1. **Not essential for basic operation** — A computer can function (process data, run programs) without a modem
2. **External connectivity** — It connects to the computer via USB, Ethernet, or PCIe slot
3. **Input/Output role** — It acts as both an **input device** (receiving data from internet) and **output device** (sending data to internet)
4. **Removable/Replaceable** — Can be added, removed, or upgraded independently
5. **Auxiliary function** — Its purpose (internet connectivity) is supplementary to the computer's core processing tasks

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Quick Revision Checklist

☐ **Q1:** Primary vs Secondary Memory, RAM (SRAM vs DRAM), ROM types, Cache levels, Memory Hierarchy

☐ **Q2(a)+2(b)+4:** Types of Computers (by size/purpose/data), Social Impact (positive & negative)

☐ **Q2(a)+2(b)+4:** Dec→Bin, Oct→Hex, Bin→Oct, Hex→Dec conversions

☐ **Q2(a)+2(b)+4:** Oct↔Hex via BCD (binary intermediate)

☐ **Q2(a)+2(b)+4:** Binary Multiplication and Division (long method)

☐ **Q2(a)+2(b)+4:** Binary Subtraction with 2's Complement (both positive and negative results)

☐ **Q3:** Keyboard layouts (QWERTY, DVORAK, AZERTY) and sections

☐ **Q3:** Mouse types (Mechanical, Optical, Laser, Wireless, Trackball, Gaming)

☐ **Q3:** Monitor types (CRT, LCD, LED, OLED) and comparison

☐ **Q3:** Modem — purpose, types, and why it is a peripheral